From chaos to cash

How hybrid by design creates business value

The generative AI landscape is no longer a playground for early adopters, it's a full-blown business transformation race. This isn't just an opportunity to better deploy gen AI at scale. It's about using gen AI as impetus to rethink how technology can support the business. C-suite leaders are effectively placing a business wager on their readiness to deliver and scale AI in their core operations. With the actions they take now, they're betting the business, whether they realize it or not.

This is the first in a series of reports on how to design and implement your organization's "Great Tech Reset" using a method we call hybrid by design—an architectural framework running on hybrid cloud.

IBM Institute for Business Value

IBM

This is the first in a series of reports on how to design and implement your organization's "Great Tech Reset," using a method we call hybrid by design. In this first installment, we lay out the big picture of hybrid by design—what it is, why it works, and first steps to take. Subsequent reports will cover topics from designing for business performance, to operating model and architecture decisions that lead to business value.

From chaos to cash

Using an approach we call hybrid by design, organizations can rethink their technology estate, so it not only best supports gen AI, but also their businesses moving forward.

Our research shows most organizations have low expectations for ROI on IT projects. They're being funded even if the business case promises only a 10% ROI.[1] The bar should be higher, especially for projects involving transformational technologies such as gen AI.

For decades, enterprises have tried to convert new technology into better business performance. Most only deliver value in pockets. Chasing the "Next Big Thing" in technology has left a trail of sluggish implementations and mountains of technical debt. (Tech debt occurs when today's technologies become a liability rather than an asset down the road. This usually happens when, despite sunk costs, organizations find their technology is too rigid and too unintegrated to accommodate new business objectives.)

Now, though, generative AI looms large and it's driving a technology reset—the Great Tech Reset. It's a chance to rethink technology estates, modernizing, to get the most from a symphony of technologies—from gen AI, to cloud, to platforms and software.

C-suite leaders not only have an opportunity to better deploy AI at scale, they also can use its transformative potential as a catalyst to completely overhaul their IT estates and the business outcomes their teams are capable of creating.

What is hybrid by design and how can it help the business?

Hybrid by design is a tested, codified architectural framework, running on hybrid cloud, to help organizations optimize business value through technology. It's a structure that will support their business moving forward, giving them the agility, speed, and integration they need to achieve their business outcomes in the future.

Hybrid by design began in cloud architecture. It described how some organizations designed their hybrid cloud estates—deliberately, through the lens of business priorities. These enterprises used a mix of public and private clouds, as well as on-premises data centers, to help them gain agility and speed, and scale business initiatives.

Today, as generative AI invades businesses, the principles behind hybrid by design apply beyond cloud computing to the enterprise as a whole: platforms, security, AI, cloud, data—the entire technology estate. Hybrid by design can make a cacophony of disparate technologies into a symphony—amplifying business outcomes through wise design and integration.

The norm: Hybrid by default

Instead of a tech symphony, however, the alternative and more common state for many organizations is "hybrid by default," best illustrated by an organic mix of clouds and on-premises data centers. Hybrid-by-default estates are unintentionally heterogenous, complex, siloed environments that create higher costs, lower returns, failed implementations, and buyer's remorse. They lack an overarching plan aimed at the best integrated business outcomes and are littered with technical debt.

Readiness to deliver and scale gen AI and other evolving technologies rests on an intentional, well-designed reset, with hybrid-by-design principles lighting the path forward.

Hybrid by default vs. hybrid by design

The hybrid-by-default problem	The hybrid-by-design solution
Tangled architecture and operations stifle transformational technologies.	Intentional transformation: A framework for architecture and operations powers enterprise-scale results.
Tech sprawl reigns. Investments are equally as scattered, delivering weak returns.	Intentional investment: Product roadmaps drive wise, targeted investments, creating consistent business value.
Passive, back-office tech architecture impedes enterprise-scale solutions.	Intentional architecture: Tech architecture integrates data, applications, and cloud estates.
A revolving door of vendors has no accountability for business outcomes.	Intentional partnerships: Strategic partners combine resources for shared success.
Siloes handoffs and weak governance undermine operations.	Intentional operations: Teams design end-to-end value streams to deliver better products, faster.

Hybrid by design in action

Hybrid by design has multiple benefits and can ultimately deliver all of them—speed, agility, a better customer experience, productivity—the list goes on. Most organizations will realize hybrid by design benefits over time, rather than all at once, as they reset their tech estate. Here are two organizations that are using hybrid by design to better serve their customers and clients, bringing real-world benefits through a series of step-changes.

Delta transforms with hybrid by design[2]

Challenge

When the COVID-19 pandemic began, Delta needed to respond to the economic realities of the time. Once the pandemic came to an end, it would need to quickly get new, premium customer experiences off the ground. Both directions meant that a transformation was required to enable technology solutions to be delivered faster, and in a more secure, reliable, and scalable way.

Solution

Delta mapped the migration that would move most of the airline's distributed workloads to the hybrid cloud. By modernizing operations with an open hybrid-cloud architecture, Delta can now deploy virtually anywhere and take a consistent, standards-based approach to development, security, and operations across clouds.

Outcome

Hundreds of applications have been migrated to the cloud. Delta and its customers are already reaping some of the rewards such as free in-flight wifi on more than 680 planes. The airline's cloud transformation initiative aims to increase employee engagement, productivity, speed to market, and cost efficiencies by between 25% and 30%.

Argentina's Ministry of Health (AMoH) uses hybrid by design to create a more stable IT infrastructure[3]

Challenge

Patients in Argentina often have a primary hospital for the majority of medical needs, but they also use other health facilities. Or they shuffle between private practices and public hospitals for tests. Because of this, Argentina's Ministry of Health wanted to automate the flow of public health statistics and the management of underlying systems.

Solution

Moving beyond its slow legacy solutions and monolithic applications, AMoH built a national digital health network. The ministry established a flexible yet stable IT infrastructure with a Red Hat® technology foundation, allowing centers to securely access patient data through standardized integration between providers.

Outcome

The new digital health network allowed AMoH to respond to a 1,500% increase in transaction volume during the COVID-19 pandemic.

It also allowed the institution to manage universal electronic records, enabling a quicker response to increases in transaction volume, while also allowing the addition of new services and features.

The IBM Institute for Business Value has identified three things leaders need to know and three things they need to do to get started on hybrid by design.

	What to know	What to do
01	A tech reset now equals durable competitive advantage later.	**Raise the bar with a reset.** For example, if you're currently delivering 20% ROI on 20% of the IT budget, reset it a step-change higher.
02	Generative AI reveals whose tech estate is built on sand and whose is built on solid ground.	**Rebuild the foundation.** **Apply hybrid-by-design architectural principles to help deliver applications at enterprise scale and convert constraints into assets.**
03	Slowing down to take stock actually speeds up your reset.	**Clear a path for enterprise-scale solutions.** **Tackle legacy technology to remove obstacles to a new way of working and operating.**

In short, a tech reset can build durable competitive advantage, but first you need to look at your technology estate through a gen AI lens. Is it set up to support gen AI workflows, powered by hybrid cloud? You will likely find that tech debt is getting in the way and will hinder the value achieved until it's addressed. Simply put, to accelerate and scale the impact of data and AI across the enterprise—and ultimately improve business outcomes—organizations must be hybrid by design.

What you need to know:

01. A tech reset today equals durable advantage tomorrow

Cutting-edge technology should not dull the bottom line. 72% of executives agree that improving ROI from the IT investment portfolio by 25% or more is a top C-suite priority for 2024.[4]

But to do that, leaders need to design their hybrid model in a way that frontloads value, an antidote to the "dabbling-level" adoption that is so common and leads to weak ROI.

For instance, almost one-third of organizations say their cloud journey stalled midway.[5] And a further 37% report being "done" after only minimal workload migration.[6] Too often, cloud loses momentum before investments start to pay off. Dabbling-level adoption—a project here and there—stops short of a tipping point where the ROI from improvements in business performance balances, and then outpaces, implementation costs. Consequently, cloud programs can be seen as necessary but unwelcome drains on resources, rather than opportunities for reinvention.

Only 29%

of cloud IT assets and services are performing as required.

Dabbling is not just limited to cloud, though. 55% of executives report that designing IT solutions to solve critical business challenges is a significant obstacle or a real roadblock.[7] To help ensure the approach to generative AI doesn't follow in the footsteps of previous unsuccessful approaches, a tech reset is necessary. Currently, only 29% of cloud IT assets and services are performing as required.[8] The remaining 71% is essentially tech debt.

Scaled gen AI: Raising the bar for ROI

Executives have low expectations even for IT projects explicitly designed to improve business outcomes, according to our latest research. Other data shows that some organizations are getting better overall at converting digital initiatives into business value; 24% of respondents claimed two times the returns on at least some investments in 2022 and 35% expect to do so in 2024.[9] The bar can be higher. We'll discuss this in more detail in our upcoming report on the economics of hybrid by design.

It's not about spending more. It's about spending better.

Leaders across business units and IT need to agree on three numbers:

1. The percentage of the IT budget available for investments to improve business performance
2. The current ROI from the entire IT portfolio
3. The length of time required to turn ideas into cash (design and delivery velocity)

These three numbers show the outcome of the way IT fuels the business today. They are strong predictors of the outcomes your organization will get with gen AI. As economist and statistician W. Edwards Deming said, "Every system is perfectly designed to get the result that it does." Different results require changes in design. Pushing new technology into the same operating model won't deliver transformational results.

How does it work in the real world? See IBM's own story of business transformation (on page 20) with a hybrid-by-design approach. The organization is en route to $3 billion in productivity improvements by the end of 2024.

72%

of executives agree that improving ROI from the IT investment portfolio by 25% or more is a top C-suite priority for 2024.

What you need to do:

01. Raise the bar with a reset.

Rethink ROI.

Durable competitive advantages aren't built on tweaks. Hybrid by design fuels continuous improvement, but aim for a moonshot. This deliberate approach seeks deep shifts, not baby steps, exposing the behavioral changes needed to unlock AI's true power.

Double down on ROI by setting a higher target for new tech efforts.

Some enterprises are already expecting to deliver over 20% ROI on some investments in 2025, so the target should be north of 30% or even higher.[10] Improving the early stages of program design, upgrading digital product engineering capabilities, and modernizing IT investments can close the gap.

Shift IT budget focus from keeping the lights on to powering breakthrough solutions.

Make more of the IT budget available for investments that can solve critical business problems such as productivity lag and revenue slumps. The shape of a typical IT investment portfolio leaves only the tip of the pyramid available to improve business performance. Absent significant increases in the IT budget, making more than 20% or so of the budget available means building new platforms, working with partners to reduce maintenance costs, modernizing legacy assets, and eliminating tech debt.[11]

Codify to simplify . . .

. . . and simplify to gain speed, reducing the lead time required to convert IT ideas into business outcomes. Hybrid by design improves velocity by shortening the time it takes to deliver digital products. It codifies architectural technology decisions that help developers build faster and more consistently, safely, and more productively, as generative AI code-writing assistants speed up many legacy processes. Improving velocity delivers direct financial benefits, builds support from sponsors by delivering earlier returns, and makes time and funding available for even more high-ROI investments.

02. Generative AI reveals whose tech estate is built on sand and whose is built on solid ground.

Intentional investments in your new hybrid-by-design technology estate form the foundation for success of your generative AI initiatives. Yet only 16% of executives say they're very confident that their cloud and data capabilities are fully ready to support generative AI investments in 2024, while 27% of executives say they're unsure of readiness.[12]

Hurdles to scaled gen AI

The path from gen AI pilots to real-world deployments is often littered with roadblocks, including:

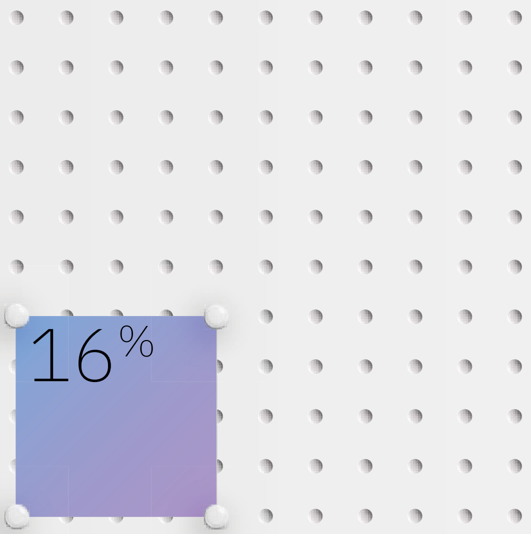
- *Barriers to free data flow.* Disparate systems create friction, meaning that workflows struggle to function across inconsistent IT stacks. A CRM system here, a marketing automation platform there often don't play well with each other. When data can't flow freely, collaboration and innovation suffer.
- *A fragmented governance structure.* Scattered and disconnected workflows can lead to shadow IT, duplicated efforts, and potential compliance issues.
- *Security concerns.* Managing security and compliance across the entire IT estate is crucial in a world where cybersecurity is increasingly essential.

Creating a gen-AI-ready tech foundation

Generative AI thrives on data. It needs vast amounts of clean, accurate information to learn and generate effective outputs—which is what enterprises need to fuel fast innovation. A robust data infrastructure—including data lakes, warehouses, and high-speed pipelines—is essential to feed the AI engine. By investing in a modern tech stack, you're not just laying the groundwork for gen AI success—you're building a foundation for consistent innovation through the free exchange of information between cross-functional teams.

Gen AI models are computationally intensive. Training and running these models require significant processing power. Legacy systems simply won't have the muscle to handle the demands of generative AI.

Becoming hybrid by design demands a comprehensive assessment of current compute capacity, data distribution (cloud, on premises, edge), data access protocols, security controls, and the potential to leverage existing technology investments. This approach doesn't just make technology more reliable (less downtime, smoother operations), it also makes an organization more adaptable (easier to react to changes, faster decision-making). Imagine seamlessly connecting everything, onsite and online, to create a perfect environment for generative AI. This translates to real results—a smarter way to work that drives profits.



Only 16% of executives say they're very confident that their cloud and data capabilities are fully ready to support generative AI investments in 2024.

What you need to do:

02. Rebuild the foundation.

Use hybrid-by-design architectural principles to pay off tech debt and convert constraints into assets.

Every trip through the Next-Big-Thing-in-IT hype cycle has added to the enterprise's store of technical debt. For instance, remember custom-built enterprise software solutions? Tailored, yes, but also incredibly expensive and time-consuming to develop and maintain. As technology evolved, these custom systems became outdated and difficult to integrate with newer tools, yet many organizations continued to maintain them. These tools and more have contributed to a convoluted technology legacy within large enterprises, so the current condition of the IT estate is not the AI-ready, enterprise-scale foundation the generative AI era needs.

Gen AI is not just the Next Big Thing in IT. This time, it is a technology that demands a shift in the fundamental way large enterprises work. Building a hybrid-by-design approach paves the way for a roadmap of improvements.

Run, don't walk, toward critical business problems.

Build the highest-impact AI products to start to create a strong foundation. Don't begin with tech use cases, though. Instead, start with critical business problems where gen AI can yield the highest return on investment. Run pilots and proofs of concept, but in areas of the business where even incremental improvements can be scaled to generate outsized returns. AI is a transformational technology; demand transformational solutions and investment cases.

“Wake up” assets to support high-ROI AI plays.

Put your dormant IT assets to good use. Consider hybrid-by-default clouds with excess capacity, data that can be freed from silos, on-prem infrastructure, mainframes that can run AI applications, and legacy applications that can be modernized as cloud-ready, AI-ready assets. Activating these dormant assets may cost money, but gen AI can support the business case for some of those investments and turn them into hybrid-by-design resources. In the short term, if a prospective AI play requires architectural changes that the business case won’t support, look for adjacent plays where the same costs can be distributed across multiple investments. Anything that doesn’t support your big plays is a form of tech debt.

Modernize smarter, not harder.

Reduce the cost of application modernization by using generative AI to build your hybrid-by-design models. Activating legacy applications puts more of the IT budget to work, but the cost of doing so has often been prohibitive. Every year the same candidates for modernization come up, and every year the cost is too high. Gen AI can change the game by helping developers do the code translation and development that account for a big chunk of app modernization costs, helping reduce time-to-value.

What you need to know:

03. Slowing down to take stock actually speeds up your reset.

Enteprises can't be successful when they are rapidly dividing their resources among too many disparate initiatives. They need to take the time to identify a few key critical areas—the ones that can create the highest business value—so they can focus and scale them as quickly as possible. In other words, slowing down actually helps you move faster in the long run.

Instead, what happens far too often is that enterprises move fast but resources are diluted. Three decades of adopting the Next Big Thing in business technology have delivered expensive, uncertain business outcomes. AI adoption is poised to repeat the same pattern. Taking a moment to be intentional now can help you avoid the following:

Historically, an astounding 84% of digital transformation programs fail.[13]

And 55% of companies report technical debt as an obstacle to achieving business goals.[14]

84%

An astounding 84% of digital transformation programs fail.

Every technology revolution prompts organizational introspection, but AI's success hinges on it.

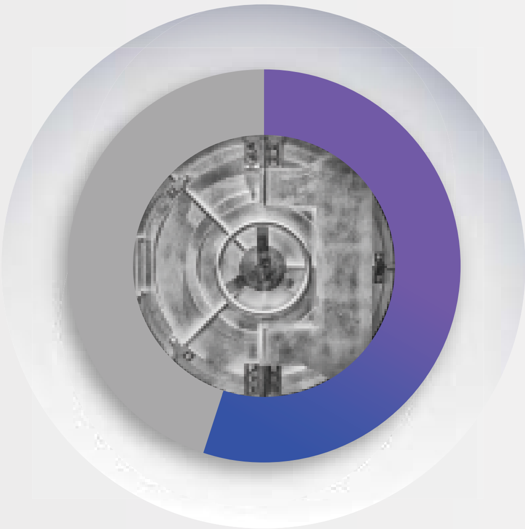
Take a moment to conduct an unflinching assessment through the rearview mirror—and then go full-steam ahead

Winning the AI game requires an unflinching business and IT assessment of the organization's readiness for the AI revolution that's underway. It's not about finger-pointing regarding the journey to the current tech estate. It's about clearly assessing the hybrid-by-default state of the organization and just as clearly mapping the benefits of hybrid by design.

When business leaders cite benefits of a hybrid-by-design approach, they include modernization, agility, security, business acceleration, cost optimization—*and the ability to unlock the power of generative AI*. In other words, gen AI can unlock *transformative* value only when it is integrated in a well-designed hybrid environment.

Exploring the path you have taken to the current tech estate can pave the way for a more deliberate, value-driven approach. Shedding the baggage of unintentional tech debt and embracing an intentional hybrid-by-design architecture designed to optimize the potential of AI is the way forward.

The good news is that the road ahead is not a secret path of the tech illuminati. It is the set of fundamental changes you know the organization needs—sort of like getting your annual health physical—but have been putting off for another day. Today is the day. Take a brief pause so you can clear a path ahead.

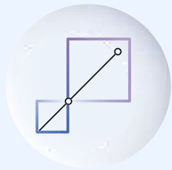


55%

of companies report technical debt either as a major obstacle or a real roadblock to achieving business goals.

What you need to do:

03. Clear a path for enterprise-scale solutions.



Tackle legacy technology to remove obstacles to a new way of working and operating.

Across business units, business functions, and IT, leaders must agree on a baseline for three critical performance metrics. Those numbers are a good indicator of the outcomes you can expect from your gen AI initiatives, if you don't implement a hybrid-by-design tech reset. Are they good enough?

Grow the portion of the IT budget that's available to improve business performance.

This isn't just about the IT budget and "shadow" spending; it's about making existing resources available to invest in AI-driven business performance improvement. The average enterprise allocates about 20% of the IT budget to invest in initiatives that improve business performance. That spend is a "good cost" because it's working to directly improve better business outcomes. It's an even better cost when it funds initiatives designed to be the very best investments the organization can imagine. Hybrid by design puts more IT assets to work delivering things customers will pay for. It converts bad costs—costs that may be necessary to the business, but that a customer would not pay for—into good costs.

Amplify the returns you’re getting from all IT spending across the entire IT portfolio.

Because enterprise IT is usually managed as a cost center, arriving at this number may be difficult and the calculation may be hard to accept. The number will likely show that the bulk of the IT budget isn’t performing as an investment. Simply cutting IT spending is one way to improve your returns, but it’s not the best way. Activating existing assets amplifies returns by putting more of the IT portfolio to work. Legacy systems, applications, and infrastructure can be modernized and reactivated. Outsourced services can provide a gen AI dividend. IT tasks can be automated. Developers can get an assist from gen AI. Platforms can consolidate application spending. The important thing is to agree—across IT and lines of business—on a baseline for tracking the ROI impact of implementing a hybrid-by-design framework.

Slash the lead time required to convert IT ideas into business outcomes.

Product-led development and digital development get the business and IT collaborating to deliver better customer and employee experiences that lead to growth and productivity. That’s good progress, but it’s not enough. Improving velocity turns investments into outcomes earlier and getting outcomes earlier boosts ROI, which creates space for more investments. Hybrid-by-design principles guide an end-to-end redesign of the concept-to-cash value stream.

$3 billion in productivity improvements

Successful implementation of AI requires more than a technology transformation. It takes an intentional hybrid-by-design approach, allowing business needs to dictate IT strategy and implementation.

It’s a $3 billion productivity opportunity.

As IBM CFO, Jim Kavanaugh, said regarding the company’s FY2023 earnings: “Against a target of $2 billion in annual run rate savings by the end of 2024, which I mentioned back in April of last year, we have already achieved over $1.5 billion. Our productivity initiatives have allowed us to increase our investments in innovation, technical and industry skills, and go-to-market capabilities, including our ecosystem. And we have accomplished this while simultaneously growing our profit margin and free cash flow, which in turn has increased our financial flexibility. This remains our playbook going forward. And given our success to date, we now believe we can achieve at least $3 billion in annual run rate savings by the end of 2024.”[15]

In the quest for greater productivity, IBM is leveraging its technology, consulting business process expertise, and strategic partnership technologies to reimagine new ways of working in a simpler IBM.

Unleashing productivity is a top priority for IBM CEO, Arvind Krishna. IBM is embedding AI into every enterprise-wide process, scaling to enable hundreds of thousands of IBMers across more than 170 countries to be more productive.

The mantra is: eliminate complexity, simplify how work is done, automate manual tasks, and embed watsonx™ everywhere. Ask first: what can we stop doing? Second: how can we simplify workflows? Then and only then, automate manual tasks with embedded AI (otherwise there's risk for automating bad processes).

Key to the strategy for workflow transformation is a greater integration of data across the enterprise—which requires a hybrid cloud strategy deliberately designed for business value. Within IBM, we are using watsonx on our hybrid cloud to infuse generative AI into business processes—putting the savings back into IBM to drive growth and investment.

50% — *reduction in cycle time, allowing...*

15% — *more time with clients*

Results include:

Reducing the average cost of running an application by

90%

Reducing the overall application environment by

50%

94%

of company-wide HR requests handled by digital assistant, AskHR

Mobilizing innovative IBMers as "productivity catalysts," who in

1,000

workshops have identified more than

5,000

grassroots opportunities

The Great Tech Reset

Moving from chaos to cash with hybrid by design

As gen AI forces organizations to assess their technical foundations, they can gain durable competitive advantage by resetting it with hybrid by design. Not only does that allow them to optimize their AI advantage, it also sets them up well for the technologies to come, which will also require agility, speed, endless capacity, and more.

In upcoming reports, we'll explore in more detail how to begin a hybrid-by-design approach, from funding, to architecture, to ecosystems, to operating model.

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Endnotes

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